

03 - Wave Equation Experiments

AC-PINN Project | Authors: Suyash Vasal Jain, Nishita Raghvendra

PDE: $u_{tt} = c^2 u_{xx}$, $c = 1.0$

IC: $u(x,0) = \sin(\pi x)$, $u_t(x,0) = 0$ | **BC:** $u(-1,t) = u(1,t) = 0$

Domain: $x \in [-1,1]$, $t \in [0,1]$

Architecture: [2, 64, 64, 64, 64, 64, 1] | **Epochs:** 10000

Note: Wave equation is second-order in time - requires velocity IC. Oscillatory nature makes it sensitive to sparse data.

```
In [1]: import sys, os
sys.path.append('.')
import torch
import numpy as np
import matplotlib.pyplot as plt
from pinn_base import (
    device, NoisyDataGenerator, PINNSolver, ACPINNSolver,
    WaveFDM, Benchmark, save_metrics, save_history, save_training_plots
)

PDE = 'wave'
LAYERS = [2, 64, 64, 64, 64, 64, 1]
EPOCHS = 10000
PDE_PARAMS = {'c': 1.0}
RESULTS = '../results/wave/'
FIGURES = '../figures/wave/'
os.makedirs(RESULTS, exist_ok=True)
os.makedirs(FIGURES, exist_ok=True)

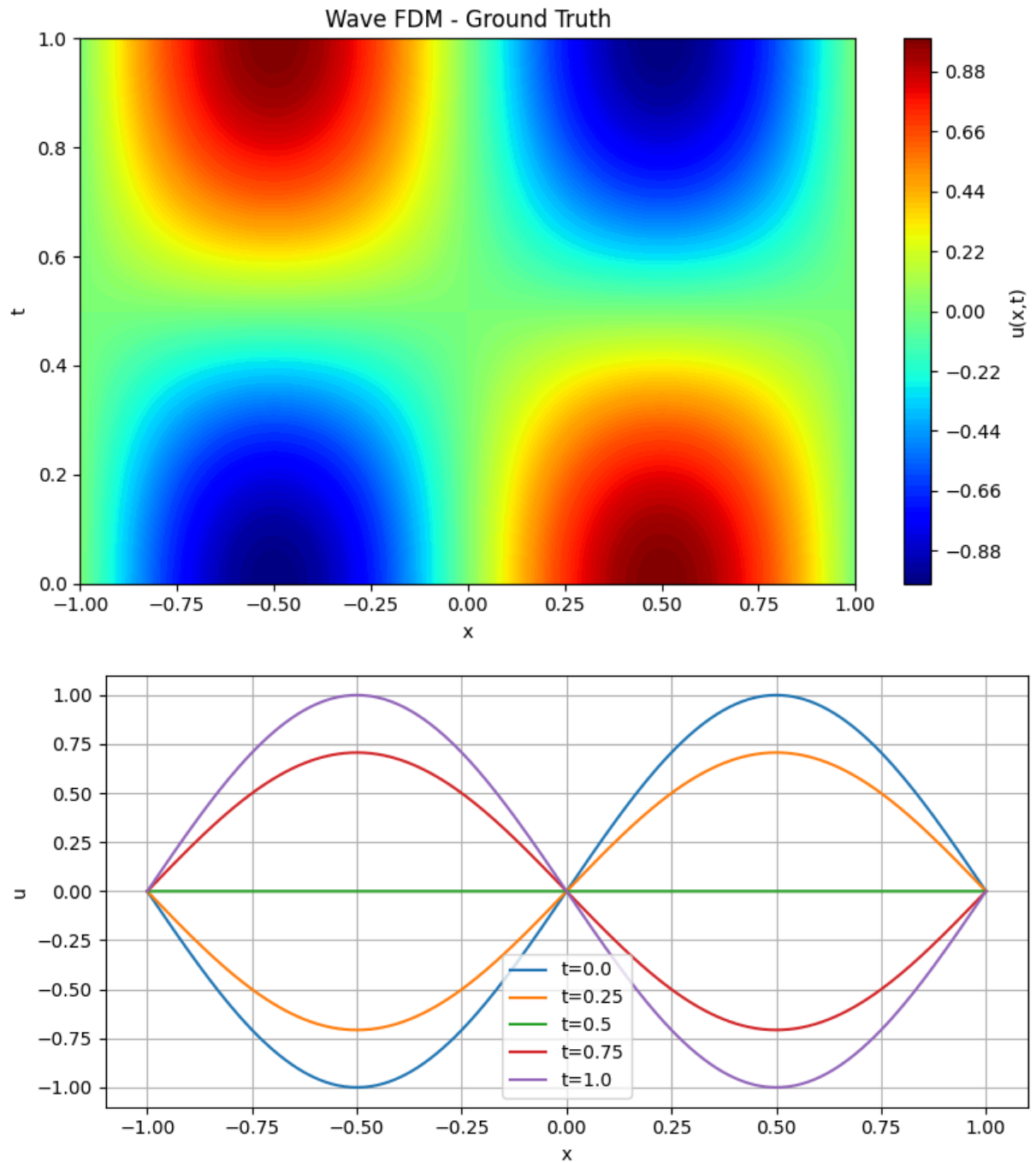
gen = NoisyDataGenerator(pde=PDE, c=1.0)
print(f'Device: {device}')
```

Device: cuda

Section 1 - FDM Ground Truth

```
In [2]: fdm = WaveFDM(nx=256, nt=2000, c=1.0)
fdm.solve()
fdm.plot_solution(title='Wave FDM - Ground Truth')
fdm.plot_time_slices()
```

WaveFDM solved in 0.0193s



Section 2 - Data Conditions

```
In [3]: data_clean_dense = gen.generate(N_ic=1000, N_bc=1000, N_f=8000, noise_eps=0.0)
data_noisy_sparse = gen.generate(N_ic=20, N_bc=20, N_f=2000, noise_eps=0.1)
print('Data ready')
print(f'Wave data includes u_ic_dt: {"u_ic_dt" in data_clean_dense}')
```

Data ready

Wave data includes u_ic_dt: True

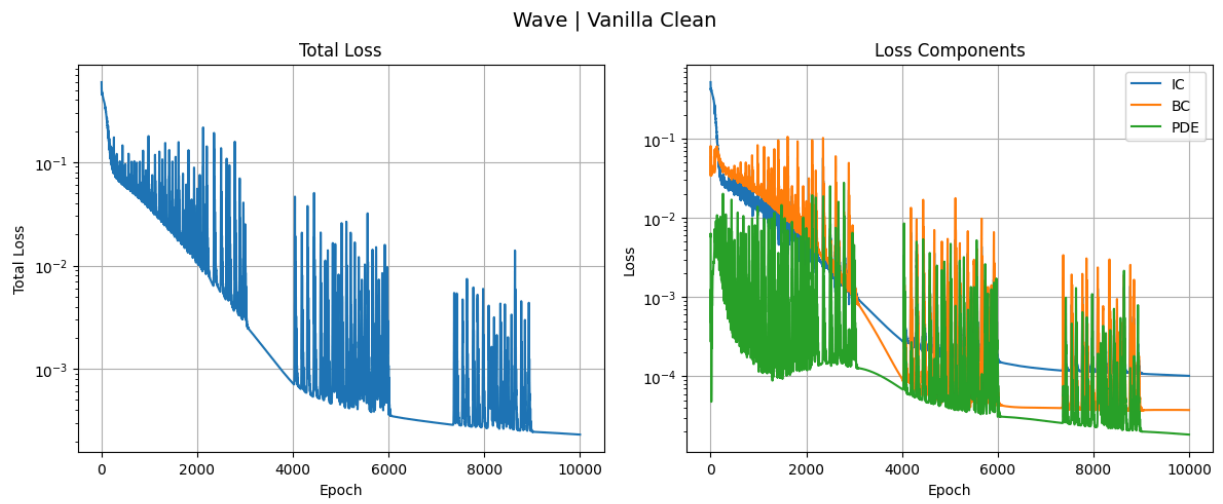
Section 3 - Vanilla PINN, Clean Dense

```
In [4]: vanilla_clean = PINNSolver(pde=PDE, layers=LAYERS, pde_params=PDE_PARAMS)
h_vc = vanilla_clean.fit(data_clean_dense, epochs=EPOCHS, print_every=1000, label='
save_training_plots(h_vc, FIGURES+'vanilla_clean_training.png', 'Wave | Vanilla Cle
vanilla_clean.plot_loss_history(h_vc)
vanilla_clean.plot_solution(title='Vanilla PINN - Clean Dense')
vanilla_clean.plot_initial_condition_comparison(gen)
save_history(h_vc, RESULTS+'vanilla_clean_history.npy')
```

D:\PINN\ac-pinn-project\ac-pinn-project\venv\Lib\site-packages\torch\autograd\graph.py:825: UserWarning: Attempting to run cuBLAS, but there was no current CUDA context! Attempting to set the primary context... (Triggered internally at C:\actions-runner\work\pytorch\pytorch\builder\windows\pytorch\aten\src\ATen\cuda\CublasHandlePool.cpp:135.)

```
return Variable._execution_engine.run_backward( # Calls into the C++ engine to run the backward pass
```

```
[Wave | Vanilla Clean | Epoch      0/10000] Total: 0.593743 | IC: 0.519431 | BC: 0.043573 | PDE: 0.006148
[Wave | Vanilla Clean | Epoch  1000/10000] Total: 0.042876 | IC: 0.019313 | BC: 0.021075 | PDE: 0.000498
[Wave | Vanilla Clean | Epoch  2000/10000] Total: 0.054513 | IC: 0.002152 | BC: 0.014973 | PDE: 0.007478
[Wave | Vanilla Clean | Epoch  3000/10000] Total: 0.025080 | IC: 0.000841 | BC: 0.001433 | PDE: 0.004561
[Wave | Vanilla Clean | Epoch  4000/10000] Total: 0.000724 | IC: 0.000283 | BC: 0.000094 | PDE: 0.000069
[Wave | Vanilla Clean | Epoch  5000/10000] Total: 0.000574 | IC: 0.000214 | BC: 0.000094 | PDE: 0.000053
[Wave | Vanilla Clean | Epoch  6000/10000] Total: 0.001076 | IC: 0.000185 | BC: 0.000074 | PDE: 0.000164
[Wave | Vanilla Clean | Epoch  7000/10000] Total: 0.000301 | IC: 0.000124 | BC: 0.000040 | PDE: 0.000027
[Wave | Vanilla Clean | Epoch  8000/10000] Total: 0.000561 | IC: 0.000118 | BC: 0.000046 | PDE: 0.000079
[Wave | Vanilla Clean | Epoch  9000/10000] Total: 0.000251 | IC: 0.000108 | BC: 0.000041 | PDE: 0.000020
Training complete in 223.00s
```

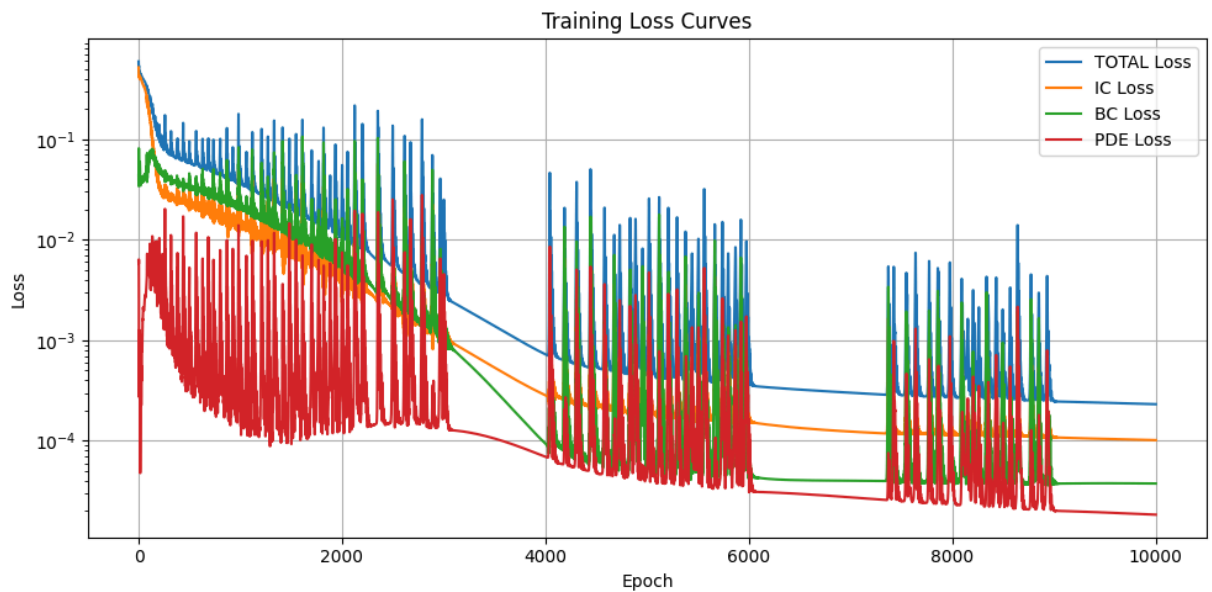


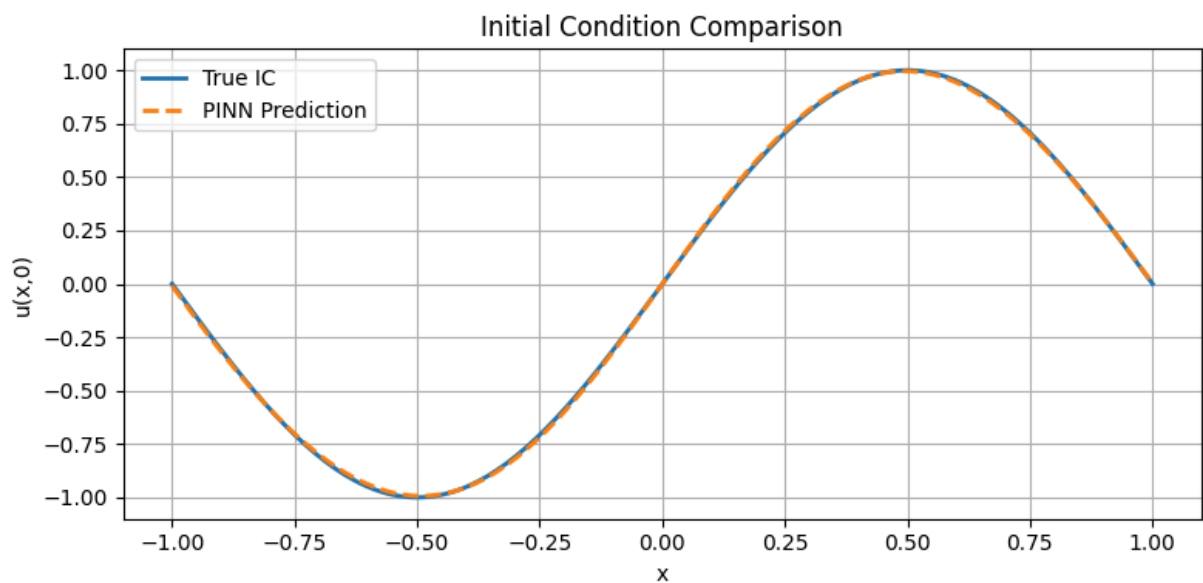
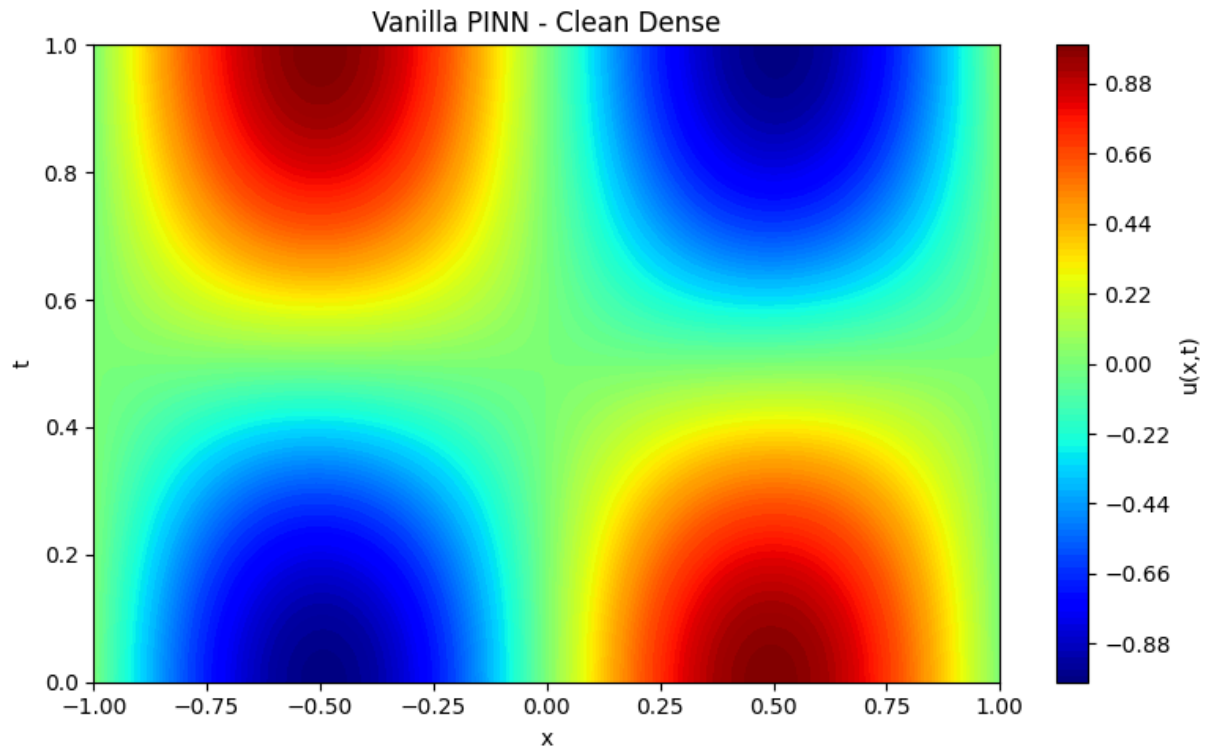
N/A - Vanilla PINN

N/A - Vanilla PINN

Saved: ../figures/wave/vanilla_clean_training.png

Saved: ../figures/wave/vanilla_clean_training.csv





Saved: ../results/wave/vanilla_clean_history.npy

Section 4 - Vanilla PINN, Noisy Sparse

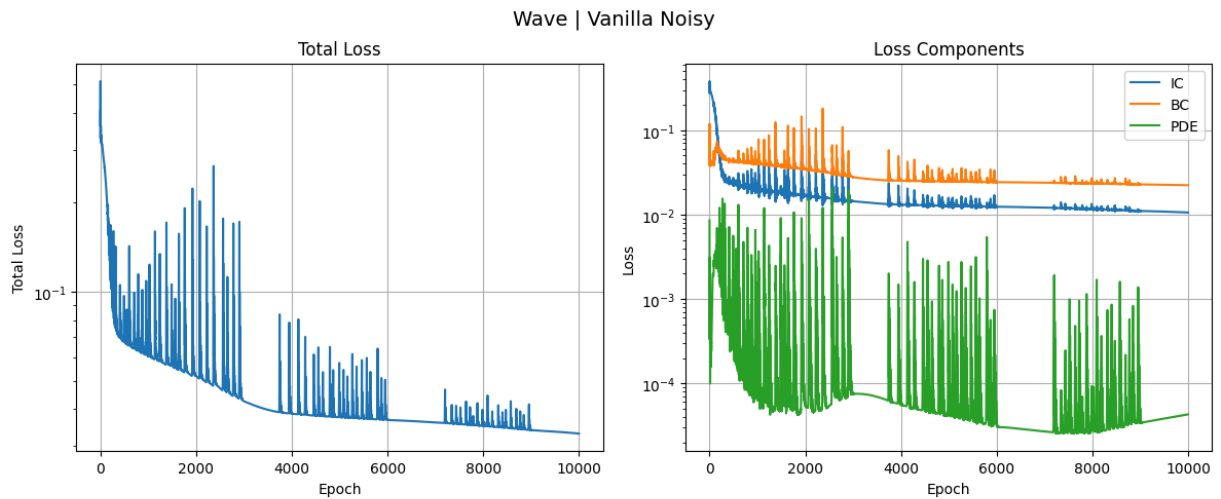
```
In [5]: vanilla_noisy = PINNSolver(pde=PDE, layers=LAYERS, pde_params=PDE_PARAMS)
h_vn = vanilla_noisy.fit(data_noisy_sparse, epochs=EPOCHS, print_every=1000, label=
save_training_plots(h_vn, FIGURES+'vanilla_noisy_training.png', 'Wave | Vanilla Noi
vanilla_noisy.plot_loss_history(h_vn)
vanilla_noisy.plot_solution(title='Vanilla PINN - Noisy Sparse')
save_history(h_vn, RESULTS+'vanilla_noisy_history.npy')
```

[Wave | Vanilla Noisy | Epoch 0/10000] Total: 0.364102 | IC: 0.283461 | BC: 0.037992 | PDE: 0.008530

```

[Wave | Vanilla Noisy | Epoch 1000/10000] Total: 0.060830 | IC: 0.019588 | BC: 0.040519 | PDE: 0.000145
[Wave | Vanilla Noisy | Epoch 2000/10000] Total: 0.051781 | IC: 0.016965 | BC: 0.034531 | PDE: 0.000057
[Wave | Vanilla Noisy | Epoch 3000/10000] Total: 0.042583 | IC: 0.014374 | BC: 0.027830 | PDE: 0.000076
[Wave | Vanilla Noisy | Epoch 4000/10000] Total: 0.038789 | IC: 0.012927 | BC: 0.025541 | PDE: 0.000064
[Wave | Vanilla Noisy | Epoch 5000/10000] Total: 0.038002 | IC: 0.012575 | BC: 0.024900 | PDE: 0.000105
[Wave | Vanilla Noisy | Epoch 6000/10000] Total: 0.036841 | IC: 0.012427 | BC: 0.024234 | PDE: 0.000036
[Wave | Vanilla Noisy | Epoch 7000/10000] Total: 0.036121 | IC: 0.012098 | BC: 0.023888 | PDE: 0.000027
[Wave | Vanilla Noisy | Epoch 8000/10000] Total: 0.035183 | IC: 0.011691 | BC: 0.023354 | PDE: 0.000028
[Wave | Vanilla Noisy | Epoch 9000/10000] Total: 0.034059 | IC: 0.010995 | BC: 0.022849 | PDE: 0.000043
Training complete in 190.31s

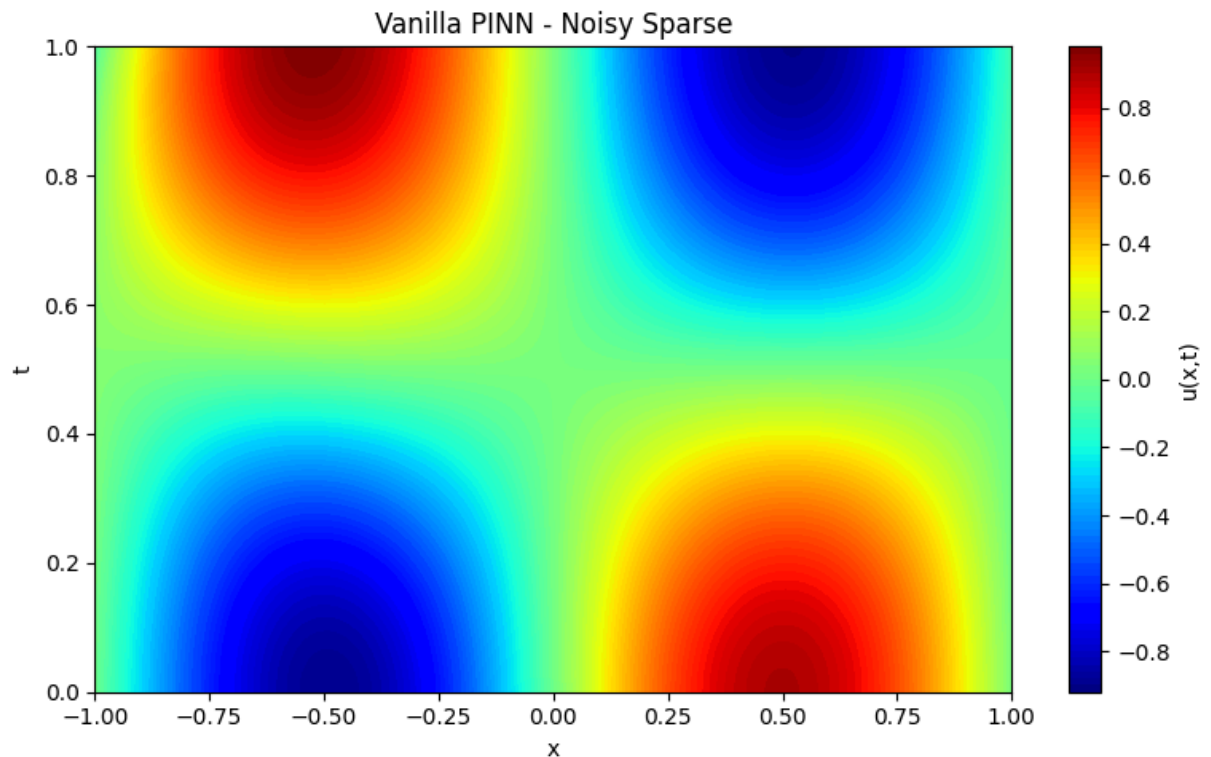
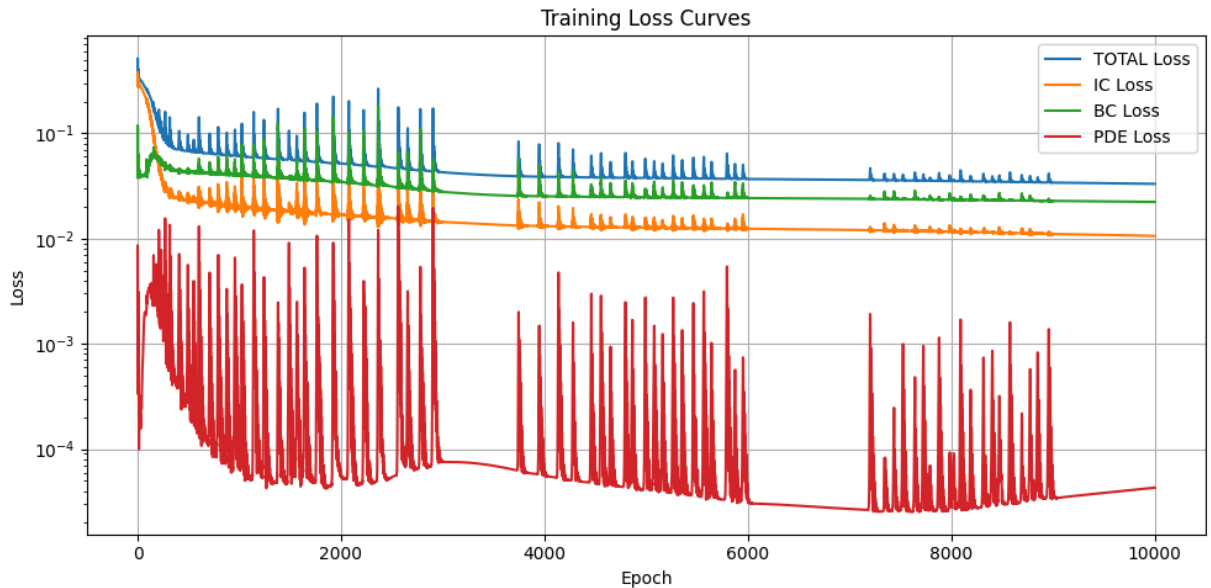
```



N/A - Vanilla PINN

N/A - Vanilla PINN

Saved: ../figures/wave/vanilla_noisy_training.png
 Saved: ../figures/wave/vanilla_noisy_training.csv



Saved: ../results/wave/vanilla_noisy_history.npy

Section 5 - AC-PINN, Clean Dense

```
In [6]: ac_clean = ACPINNSolver(pde=PDE, layers=LAYERS, pde_params=PDE_PARAMS, weight_strat
h_ac = ac_clean.fit(data_clean_dense, epochs=EPOCHS, print_every=1000, label='Wave
save_training_plots(h_ac, FIGURES+'acpinn_clean_training.png', 'Wave | AC-PINN Clea
ac_clean.plot_loss_history(h_ac)
ac_clean.plot_weight_history(h_ac)
ac_clean.plot_solution(title='AC-PINN - Clean Dense')
save_history(h_ac, RESULTS+'ac_clean_history.npy')
```

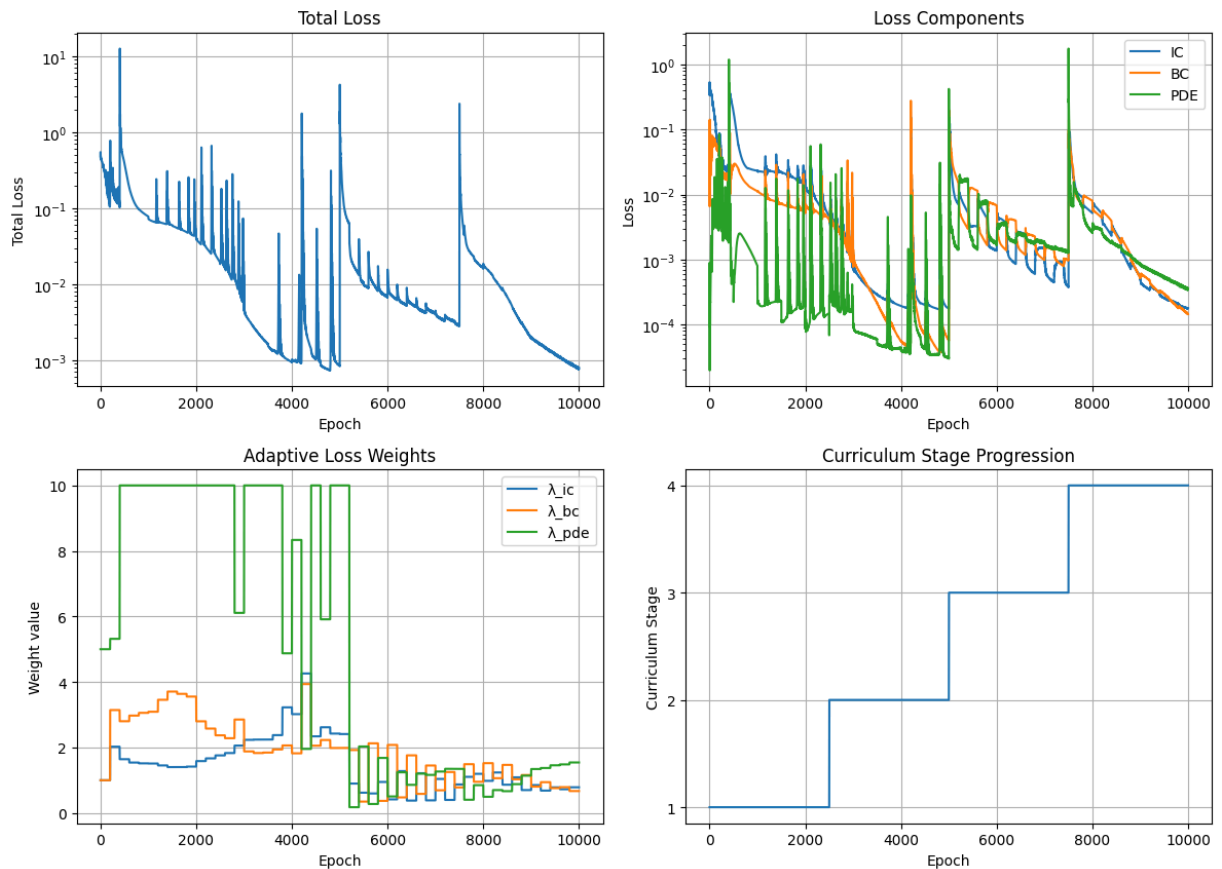
[Wave | AC-PINN Clean | Epoch 0/10000] Stage 1/4 | Total: 0.482287 | IC: 0.34614
4 | BC: 0.134824 | PDE: 0.000264 | $\lambda=(1.00,1.00,5.00)$

```

[Wave | AC-PINN Clean | Epoch 1000/10000] Stage 1/4 | Total: 0.072484 | IC: 0.02320
7 | BC: 0.011310 | PDE: 0.000245 |  $\lambda=(1.51,3.09,10.00)$ 
[Wave | AC-PINN Clean | Epoch 2000/10000] Stage 1/4 | Total: 0.039727 | IC: 0.01463
8 | BC: 0.005592 | PDE: 0.000087 |  $\lambda=(1.59,2.80,10.00)$ 
[Wave | AC-PINN Clean | Epoch 3000/10000] Stage 2/4 | Total: 0.005814 | IC: 0.00128
2 | BC: 0.001074 | PDE: 0.000093 |  $\lambda=(2.23,1.88,10.00)$ 
[Wave | AC-PINN Clean | Epoch 4000/10000] Stage 2/4 | Total: 0.001012 | IC: 0.00019
2 | BC: 0.000055 | PDE: 0.000040 |  $\lambda=(3.02,1.82,8.33)$ 
[Wave | AC-PINN Clean | Epoch 5000/10000] Stage 2/4 | Total: 0.000831 | IC: 0.00017
6 | BC: 0.000056 | PDE: 0.000030 |  $\lambda=(2.41,1.99,10.00)$ 
[Wave | AC-PINN Clean | Epoch 6000/10000] Stage 3/4 | Total: 0.015556 | IC: 0.00136
5 | BC: 0.006809 | PDE: 0.001656 |  $\lambda=(0.42,2.08,0.51)$ 
[Wave | AC-PINN Clean | Epoch 7000/10000] Stage 3/4 | Total: 0.004484 | IC: 0.00147
3 | BC: 0.000983 | PDE: 0.001793 |  $\lambda=(1.04,0.69,1.27)$ 
[Wave | AC-PINN Clean | Epoch 8000/10000] Stage 4/4 | Total: 0.018606 | IC: 0.00518
2 | BC: 0.008029 | PDE: 0.002616 |  $\lambda=(0.98,1.52,0.50)$ 
[Wave | AC-PINN Clean | Epoch 9000/10000] Stage 4/4 | Total: 0.002027 | IC: 0.00054
3 | BC: 0.000514 | PDE: 0.000857 |  $\lambda=(0.85,0.81,1.34)$ 
AC-PINN training complete in 230.41s

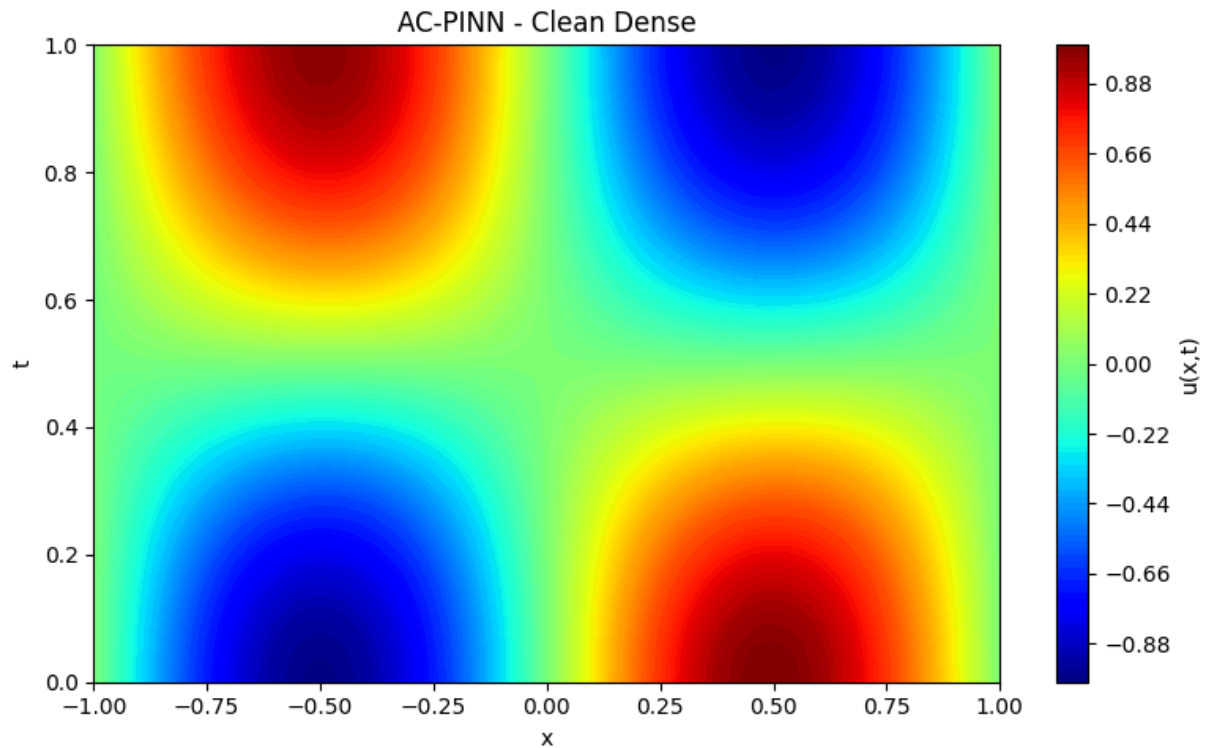
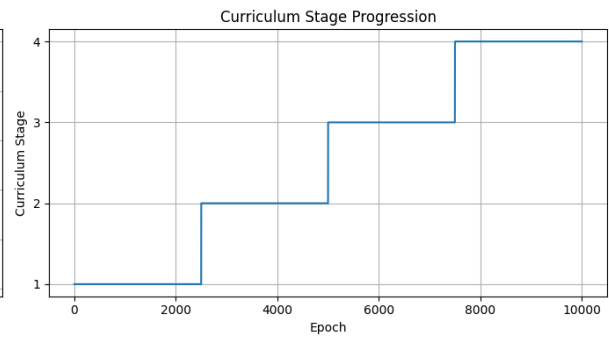
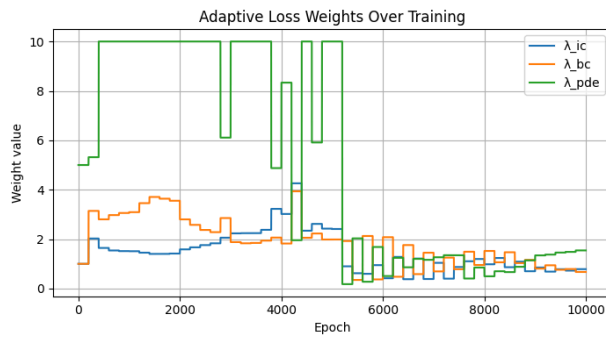
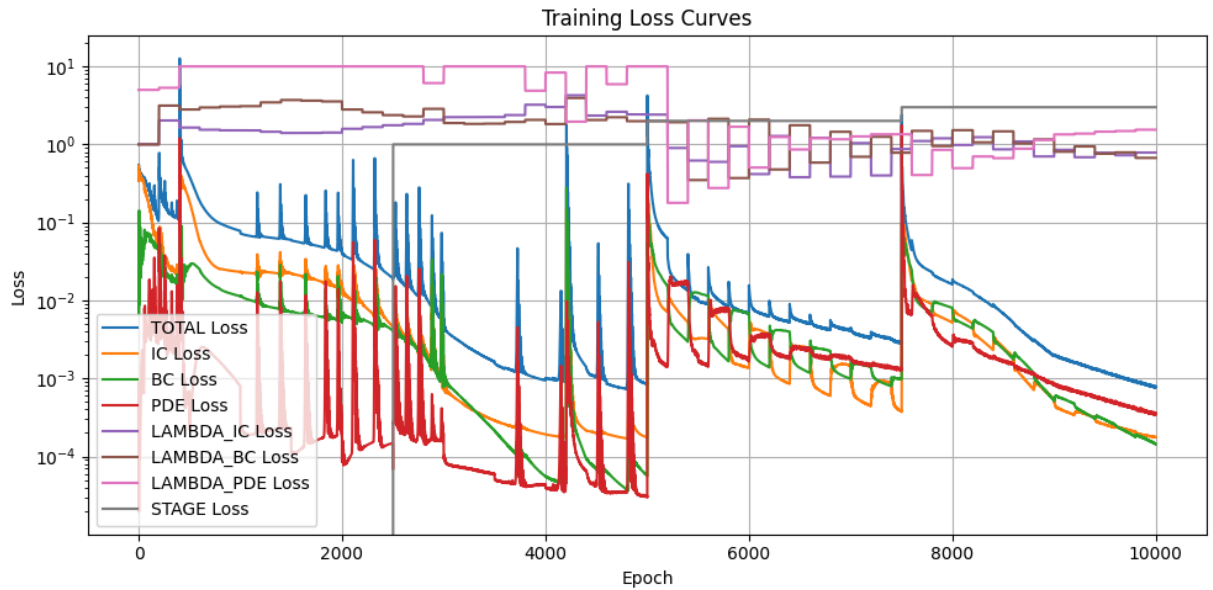
```

Wave | AC-PINN Clean



Saved: ../figures/wave/acpinn_clean_training.png

Saved: ../figures/wave/acpinn_clean_training.csv



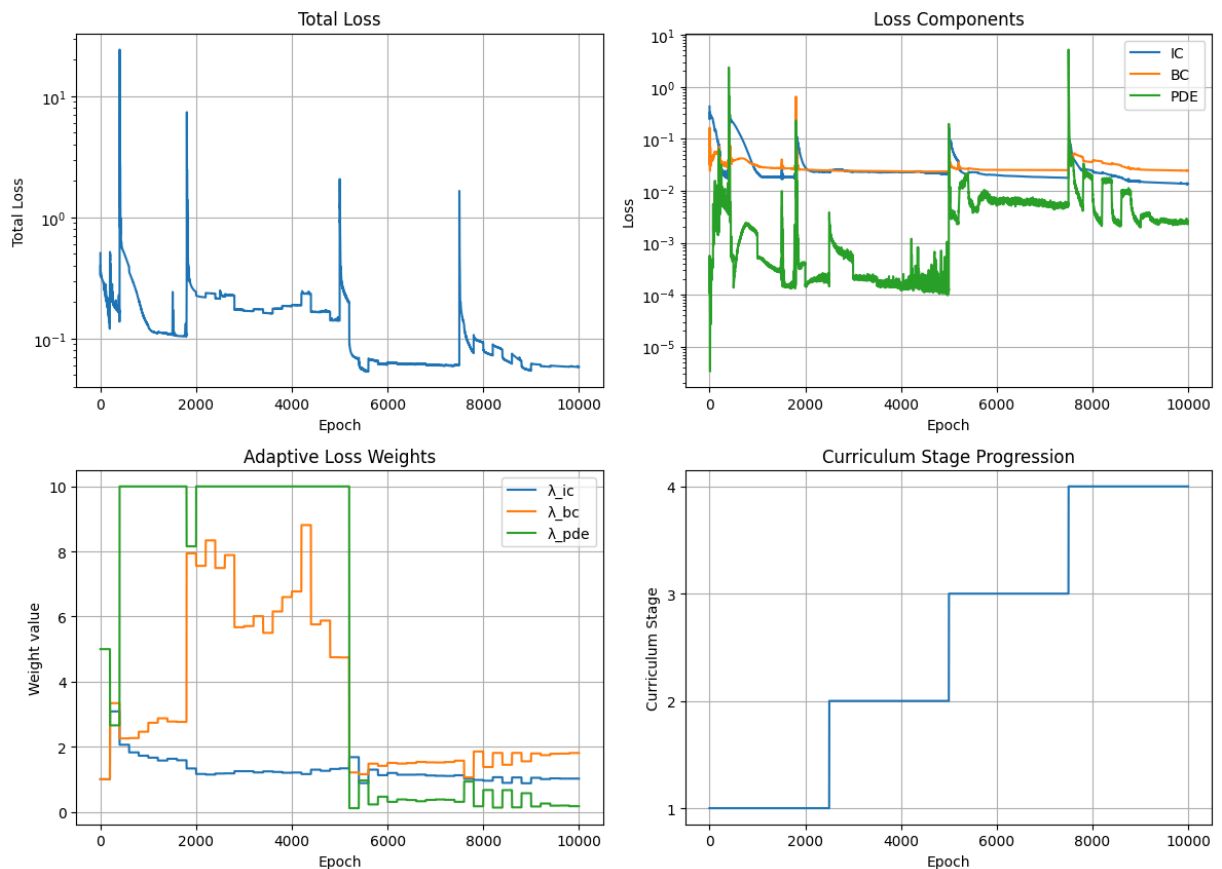
Saved: ../results/wave/ac_clean_history.npy

Section 6 - AC-PINN, Noisy Sparse

```
In [7]: ac_noisy = ACPINNSolver(pde=PDE, layers=LAYERS, pde_params=PDE_PARAMS, weight_strat
h_an = ac_noisy.fit(data_noisy_sparse, epochs=EPOCHS, print_every=1000, label='Wave
save_training_plots(h_an, FIGURES+'acpinn_noisy_training.png', 'Wave | AC-PINN Nois
ac_noisy.plot_loss_history(h_an)
ac_noisy.plot_weight_history(h_an)
ac_noisy.plot_solution(title='AC-PINN - Noisy Sparse')
save_history(h_an, RESULTS+'ac_noisy_history.npy')
```

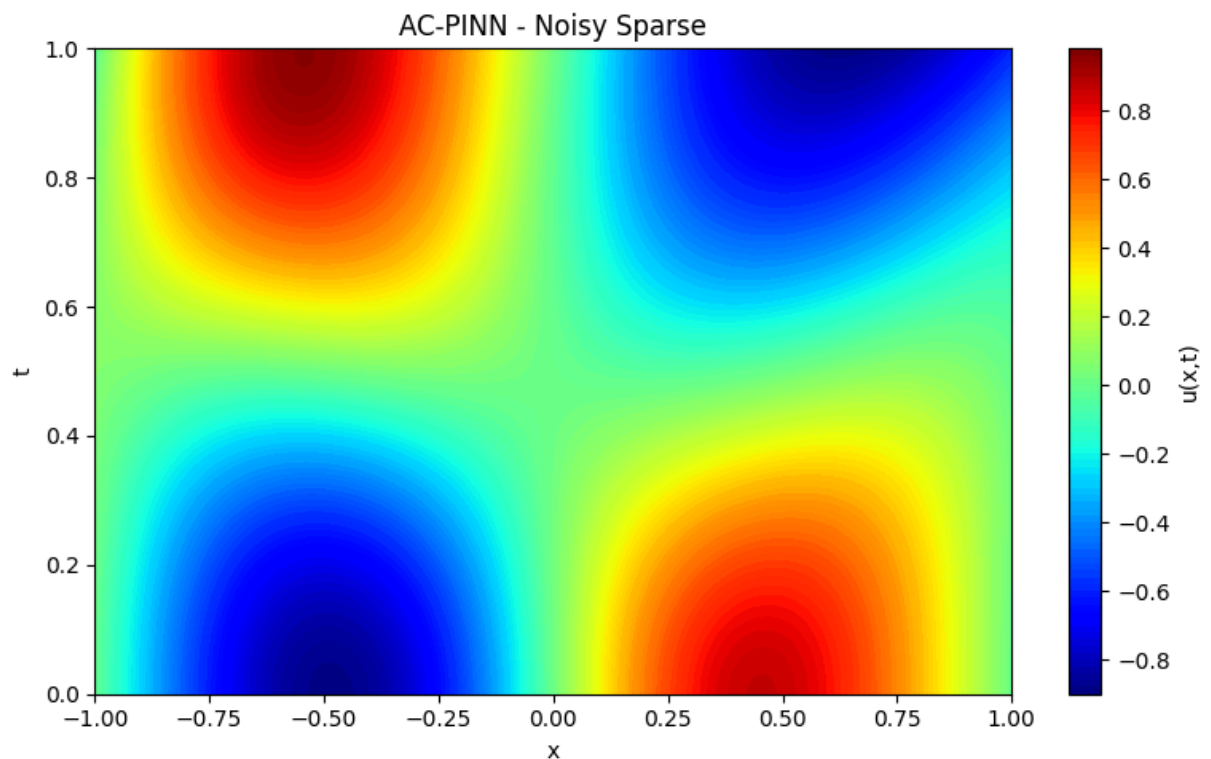
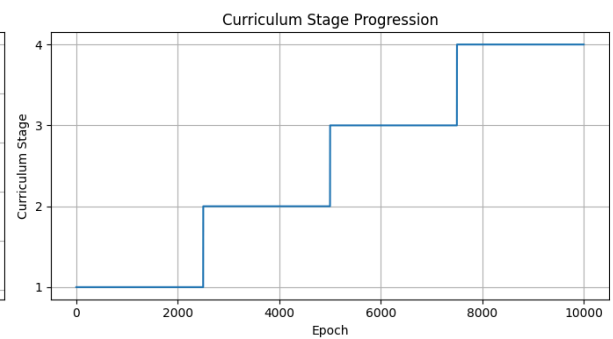
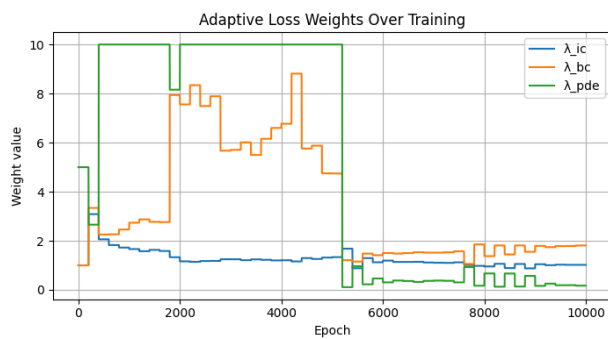
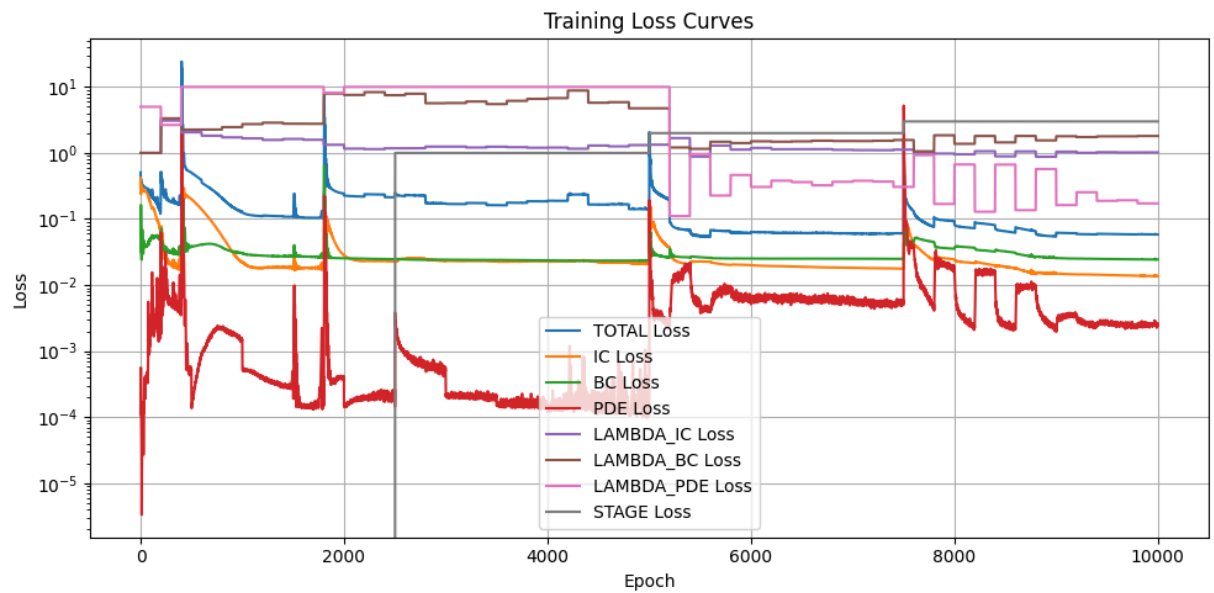
```
[Wave | AC-PINN Noisy | Epoch    0/10000] Stage 1/4 | Total: 0.507298 | IC: 0.41819
9 | BC: 0.088552 | PDE: 0.000109 |  $\lambda=(1.00,1.00,5.00)$ 
[Wave | AC-PINN Noisy | Epoch 1000/10000] Stage 1/4 | Total: 0.125454 | IC: 0.02207
0 | BC: 0.030416 | PDE: 0.000552 |  $\lambda=(1.66,2.74,10.00)$ 
[Wave | AC-PINN Noisy | Epoch 2000/10000] Stage 1/4 | Total: 0.226682 | IC: 0.02803
7 | BC: 0.025488 | PDE: 0.000145 |  $\lambda=(1.16,7.56,10.00)$ 
[Wave | AC-PINN Noisy | Epoch 3000/10000] Stage 2/4 | Total: 0.169619 | IC: 0.02350
4 | BC: 0.024194 | PDE: 0.000222 |  $\lambda=(1.25,5.71,10.00)$ 
[Wave | AC-PINN Noisy | Epoch 4000/10000] Stage 2/4 | Total: 0.189174 | IC: 0.02294
0 | BC: 0.023657 | PDE: 0.000130 |  $\lambda=(1.21,6.77,10.00)$ 
[Wave | AC-PINN Noisy | Epoch 5000/10000] Stage 2/4 | Total: 0.140837 | IC: 0.02092
7 | BC: 0.023549 | PDE: 0.000121 |  $\lambda=(1.33,4.74,10.00)$ 
[Wave | AC-PINN Noisy | Epoch 6000/10000] Stage 3/4 | Total: 0.063352 | IC: 0.02003
3 | BC: 0.025234 | PDE: 0.005142 |  $\lambda=(1.19,1.50,0.31)$ 
[Wave | AC-PINN Noisy | Epoch 7000/10000] Stage 3/4 | Total: 0.060486 | IC: 0.01826
9 | BC: 0.025023 | PDE: 0.006261 |  $\lambda=(1.11,1.51,0.38)$ 
[Wave | AC-PINN Noisy | Epoch 8000/10000] Stage 4/4 | Total: 0.083747 | IC: 0.02458
9 | BC: 0.035421 | PDE: 0.017239 |  $\lambda=(0.95,1.38,0.67)$ 
[Wave | AC-PINN Noisy | Epoch 9000/10000] Stage 4/4 | Total: 0.062046 | IC: 0.01495
9 | BC: 0.025680 | PDE: 0.002330 |  $\lambda=(1.04,1.79,0.16)$ 
AC-PINN training complete in 194.91s
```

Wave | AC-PINN Noisy



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Saved: ../figures/wave/acpinn_noisy_training.csv



Saved: ../results/wave/ac_noisy_history.npy

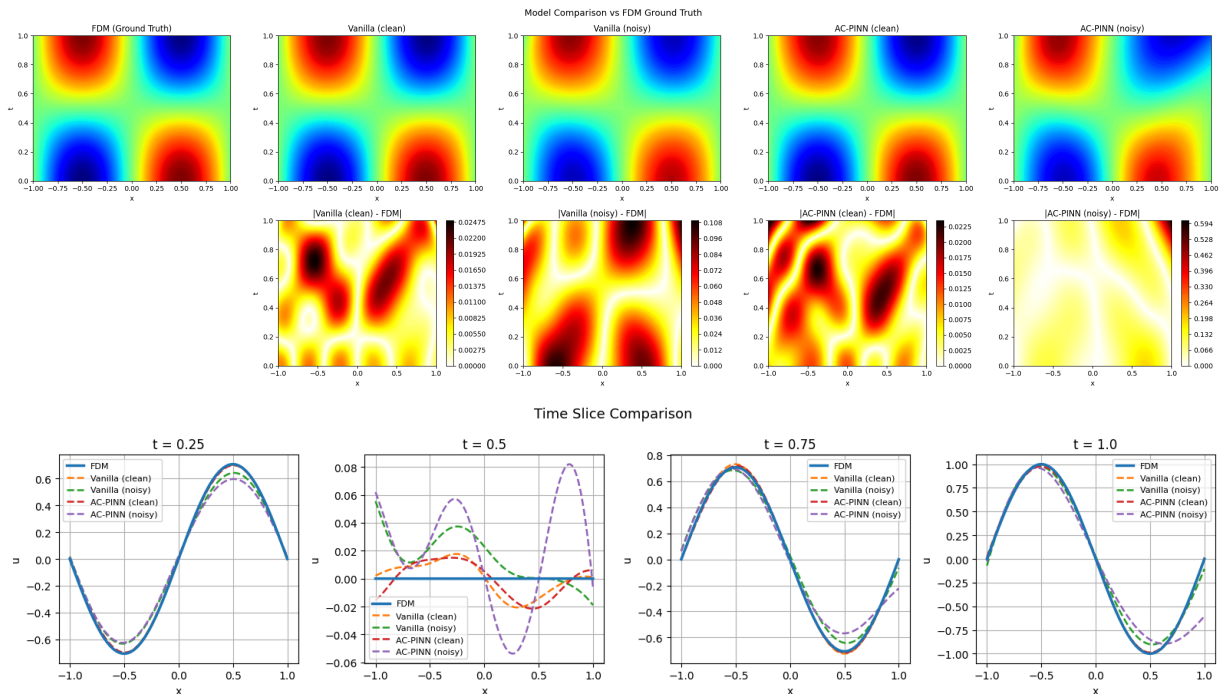
Section 7 - Benchmark vs FDM

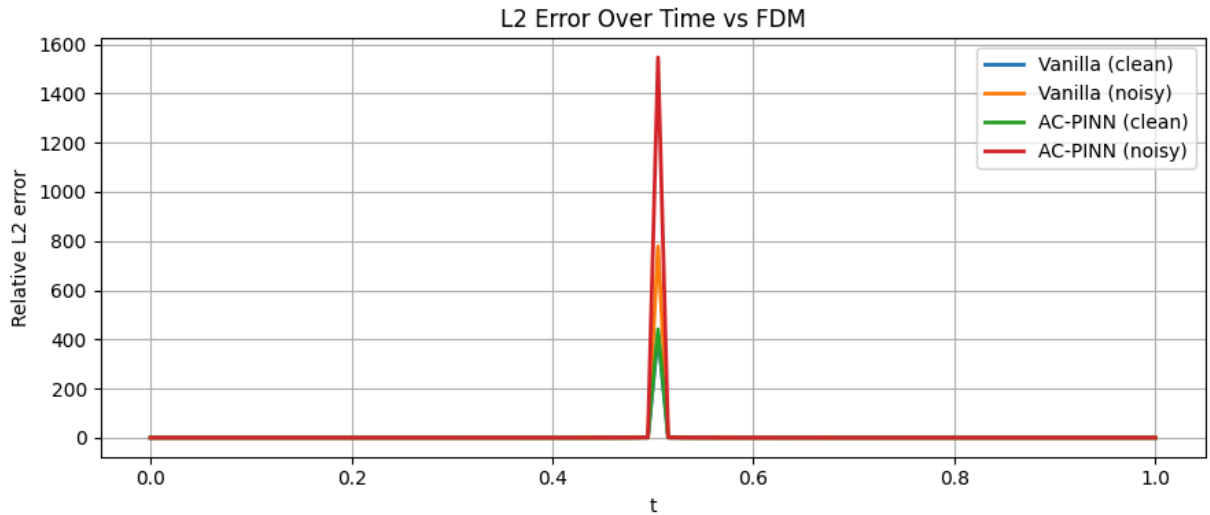
```
In [8]: bench = Benchmark(fdm, nx=200, nt=100)
bench.add('Vanilla (clean)', vanilla_clean)
bench.add('Vanilla (noisy)', vanilla_noisy)
bench.add('AC-PINN (clean)', ac_clean)
bench.add('AC-PINN (noisy)', ac_noisy)
bench.run()

metrics = bench.compare_metrics()
bench.plot_comparison(save_path=FIGURES+'comparison.png')
bench.plot_time_slices(save_path=FIGURES+'time_slices.png')
bench.plot_error_over_time(save_path=FIGURES+'error_over_time.png')

save_metrics(metrics, RESULTS+'benchmark_metrics.npy')
print('Wave experiments complete.')
```

Model	Rel L2	Max Err	MAE	RMSE
Vanilla (clean)	0.018801	0.024934	0.007343	0.009376
Vanilla (noisy)	0.096929	0.109138	0.039830	0.048340
AC-PINN (clean)	0.019802	0.023295	0.008228	0.009876
AC-PINN (noisy)	0.170908	0.603982	0.063746	0.085235





Saved: ../results/wave/benchmark_metrics.npy
Wave experiments complete.

Section 8 - Noise Level Study

```
In [9]: noise_results = {}
for eps in [0.05, 0.1, 0.2]:
    print(f'\n--- ε={eps} ---')
    d = gen.generate(N_ic=50, N_bc=50, N_f=3000, noise_eps=eps)
    v = PINNSolver(pde=PDE, layers=LAYERS, pde_params=PDE_PARAMS)
    v.fit(d, epochs=5000, print_every=2500)
    a = ACPINNSolver(pde=PDE, layers=LAYERS, pde_params=PDE_PARAMS, weight_strategy='adam')
    a.fit(d, epochs=5000, print_every=2500)
    b = Benchmark(fdm).add(f'Vanilla ε={eps}', v).add(f'AC-PINN ε={eps}', a)
    b.run()
    noise_results[eps] = b.compare_metrics()
save_metrics(noise_results, RESULTS+'noise_study_metrics.npy')
```

--- ε=0.05 ---

[Epoch 0/5000] Total: 1.634130 | IC: 1.021123 | BC: 0.511397 | PDE: 0.020322

[Epoch 2500/5000] Total: 0.045927 | IC: 0.007619 | BC: 0.009161 | PDE: 0.005830

Training complete in 95.01s

[Epoch 0/5000] Stage 1/4 | Total: 0.528315 | IC: 0.501278 | BC: 0.015112 | PDE: 0.002385 | λ=(1.00,1.00,5.00)

[Epoch 2500/5000] Stage 2/4 | Total: 0.078674 | IC: 0.019999 | BC: 0.015597 | PDE: 0.000239 | λ=(1.55,2.90,10.00)

AC-PINN training complete in 101.77s

Model	Rel L2	Max Err	MAE	RMSE
Vanilla ε=0.05	0.029391	0.052979	0.011765	0.014658
AC-PINN ε=0.05	0.067984	0.098521	0.029101	0.033905

--- ε=0.1 ---

[Epoch 0/5000] Total: 0.691021 | IC: 0.639261 | BC: 0.034849 | PDE: 0.003382

[Epoch 2500/5000] Total: 0.048586 | IC: 0.016210 | BC: 0.029103 | PDE: 0.000655

Training complete in 93.37s

[Epoch 0/5000] Stage 1/4 | Total: 0.585758 | IC: 0.538972 | BC: 0.046260 | PDE: 0.000105 | $\lambda=(1.00,1.00,5.00)$

[Epoch 2500/5000] Stage 2/4 | Total: 0.281165 | IC: 0.040171 | BC: 0.023529 | PDE: 0.000087 | $\lambda=(1.12,10.00,10.00)$

AC-PINN training complete in 100.48s

Model	Rel L2	Max Err	MAE	RMSE
Vanilla $\epsilon=0.1$	0.063150	0.118370	0.023598	0.031494
AC-PINN $\epsilon=0.1$	0.226292	0.506185	0.090444	0.112856

--- $\epsilon=0.2$ ---

[Epoch 0/5000] Total: 0.872127 | IC: 0.716694 | BC: 0.153625 | PDE: 0.000362

[Epoch 2500/5000] Total: 0.162815 | IC: 0.088676 | BC: 0.073511 | PDE: 0.000126

Training complete in 94.07s

[Epoch 0/5000] Stage 1/4 | Total: 1.067826 | IC: 0.851577 | BC: 0.213860 | PDE: 0.000478 | $\lambda=(1.00,1.00,5.00)$

[Epoch 2500/5000] Stage 2/4 | Total: 0.813094 | IC: 0.144601 | BC: 0.071204 | PDE: 0.000397 | $\lambda=(1.13,9.06,10.00)$

AC-PINN training complete in 99.78s

Model	Rel L2	Max Err	MAE	RMSE
Vanilla $\epsilon=0.2$	0.135193	0.148546	0.057627	0.067423
AC-PINN $\epsilon=0.2$	0.292573	0.355798	0.120838	0.145911

Saved: ../results/wave/noise_study_metrics.npy